

title

Teorema 1. Sean $X \neq \emptyset$, $\{(Y_\alpha, \mathcal{T}_\alpha) \text{ espacio topol\u00f3gico}\}_{\alpha \in I}$ y $\mathcal{F} = \{f_\alpha : X \rightarrow Y_\alpha\}_{\alpha \in I}$

$$\implies \sigma(X, \mathcal{F}) = \left\{ \bigcup_{\lambda \in \Lambda} \bigcap_{j=1}^{n_\lambda} f_{\alpha_{j,\lambda}}^{-1}(U_{j,\lambda}) : \forall \lambda \in \Lambda : n_\lambda \in \mathbb{N} \wedge \forall j \in \mathbb{N}_{n_\lambda} : \alpha_{j,\lambda} \in I \wedge U_{j,\lambda} \in \mathcal{T}_{\alpha_{j,\lambda}} \right\}.$$